

Immune System 1 & 2

1. Introduction
2. Organs & tissues
3. Nonspecific immunity / innate (*born with it*)
4. Overview of cells & chemicals (*mediators*)
5. Specific immunity / adaptive (*acquired*)
 - *Introduction*
 - *2 branches*
 1. *Humoral mediated – proteins in the blood (a humor) mark pathogens*
 2. *Cell mediated – cells attack pathogens*
6. Immunity & health

How pathogens/cancer create illness?

Bacteria:

- Toxins in membranes (*endotoxin*) → illness
- Toxins released (*exotoxin*) → illness

Virus:

- Invade “self” cells, take over, multiply, lyse “self” cell → illness

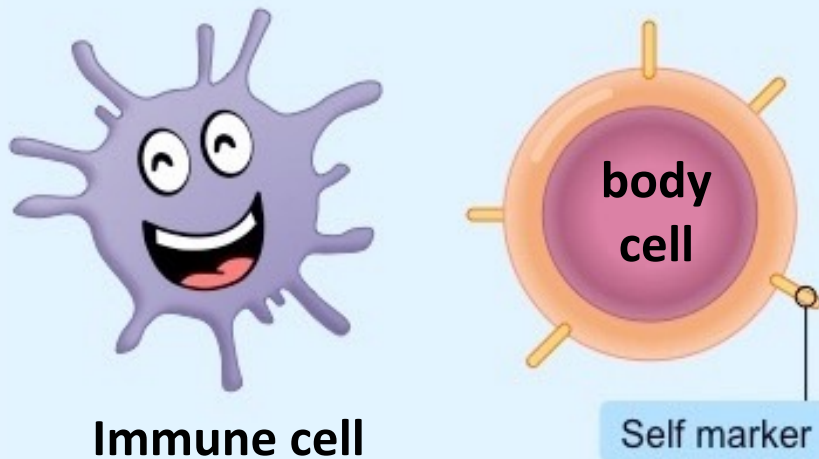
Cancer:

- Genetically altered “self” cell, overgrows normal tissue, lose normal tissue function → illness

How do immune cells distinguish “self” from foreign Ag’s?

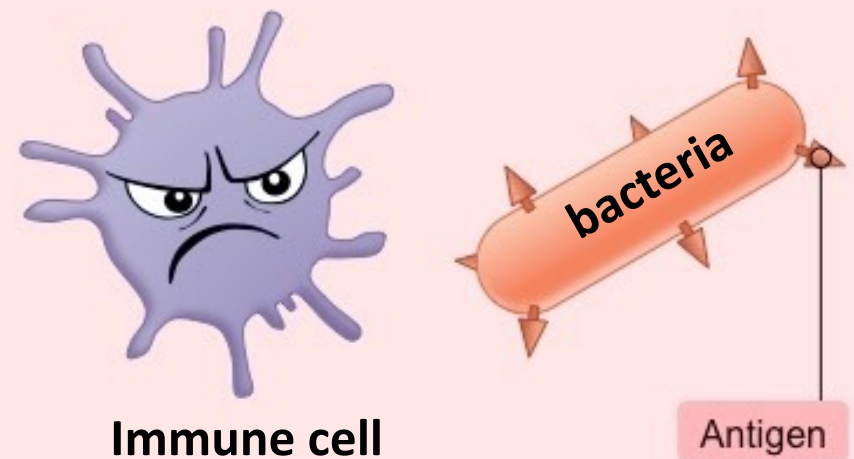
MHC surface markers on “self” cells

IDENTIFYING SELF



A **self marker** (MHC) labels the body's cells as a 'friend' and are tolerated by the immune system

IDENTIFYING NON-SELF



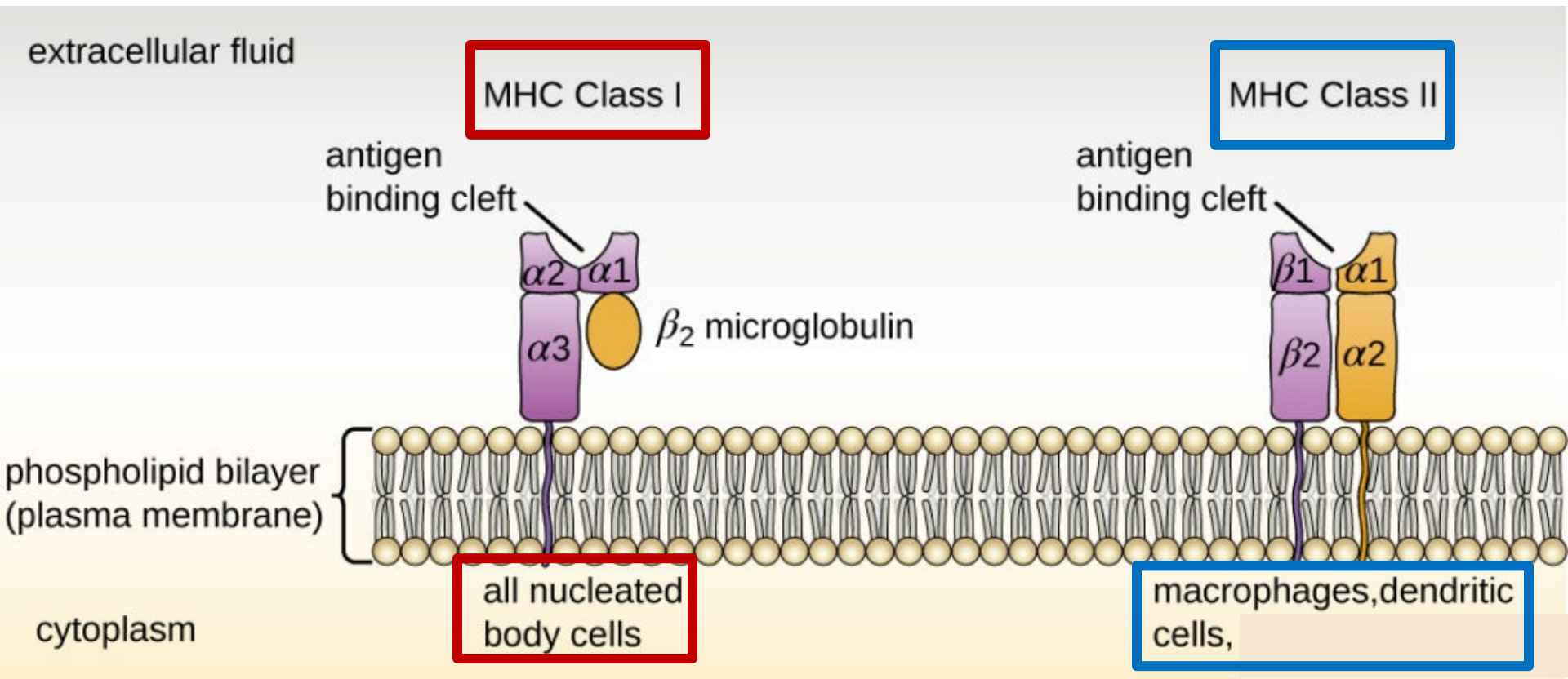
An **antigen** is a molecule that the immune system recognises as foreign (non-self) and treats as a 'foe'

MHC

- **Major Histocompatibility Complex** (*refers to a gene locus*)
- Codes for “self” surface proteins; includes MHC 1 & MHC 2

MHC Surface Markers

MHC I and *MHC II*



Not RBCs

**Antigen
presenting cells**

How are immune incompetent cells removed?

prevents autoimmunity

1. In **bone marrow** (*B cells*) & **thymus** (*T cells*)
 - **Clonal deletion**: lymphocytes that attack “self”
Ag’s are destroyed by **apoptosis** (*programmed cell death = cell suicide*)
2. In **lymphatic organs** & **tissues**
 - **Clonal inactivation**: if lymphocytes create antigen receptors to “self”, they self inactivate

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Specific Immune System

overview

Effector cells:

- *B lymphocytes*
- *T lymphocytes (helper T+ cytotoxic T cells)*

Hallmark features of specific immune response:

- ***Specificity*** – *lymphocyte clones recognize Ag's unique to each pathogen*
- ***Diversity*** - *potential to recognize ALL foreign Ag's*
- ***Distinguish self vs. non-self*** – *B/T lymphocytes must be “immunocompetent”*
- ***Memory*** – *after infection, immunity to that Ag remains through lymphocyte “memory” cells*

Specific Immune System

overview

General Process:

1. **Recognize** the pathogen
2. **Activate** the immune mechanisms
3. **Respond** to the pathogen

2 branches of the **general process**:

1. **Humoral mediated immunity**

B cells: *secrete antibodies* → *coat* pathogen → *mark for destruction*

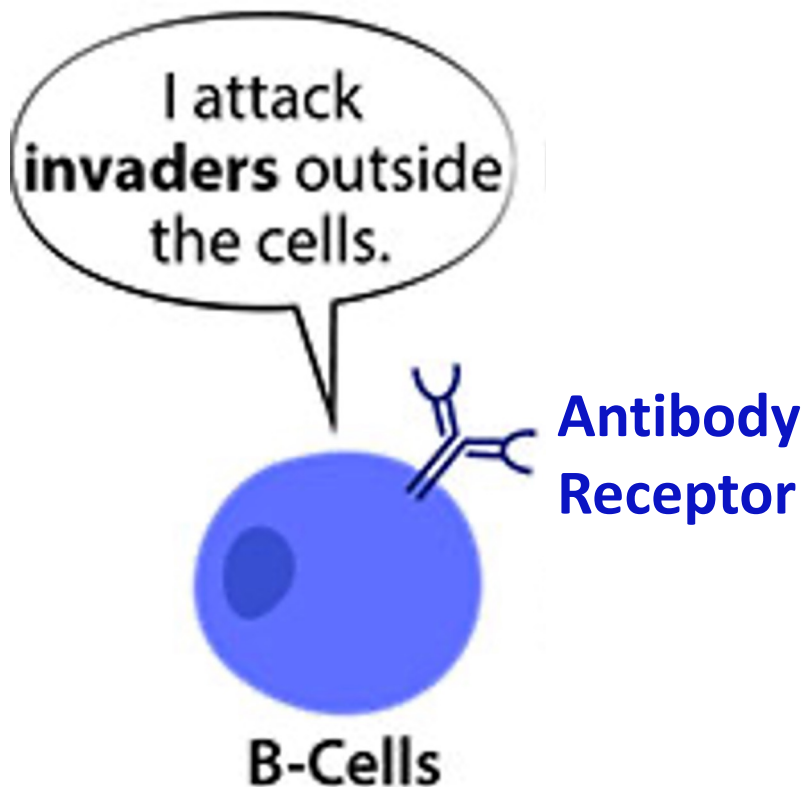
2. **Cell mediated immunity**

T cytotoxic cells: *directly contact* → *kill* pathogen

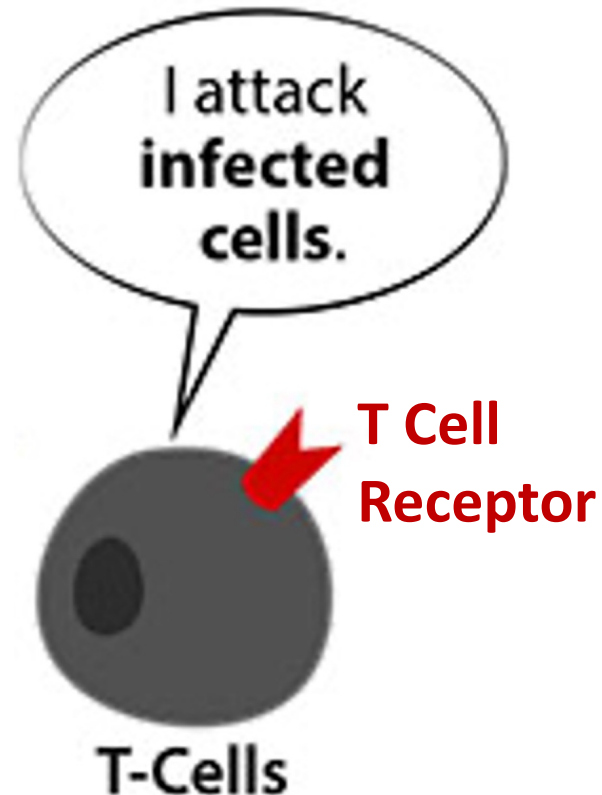
How do T & B lymphocyte recognize foreign Ag's?

Surface receptors that bind specific foreign Ag's.

B cells have **Antibody Receptors**: *recognize bacteria & free virus*



T cells have **T Cell Receptors**: *recognizes cancer cells & virally infected cells*



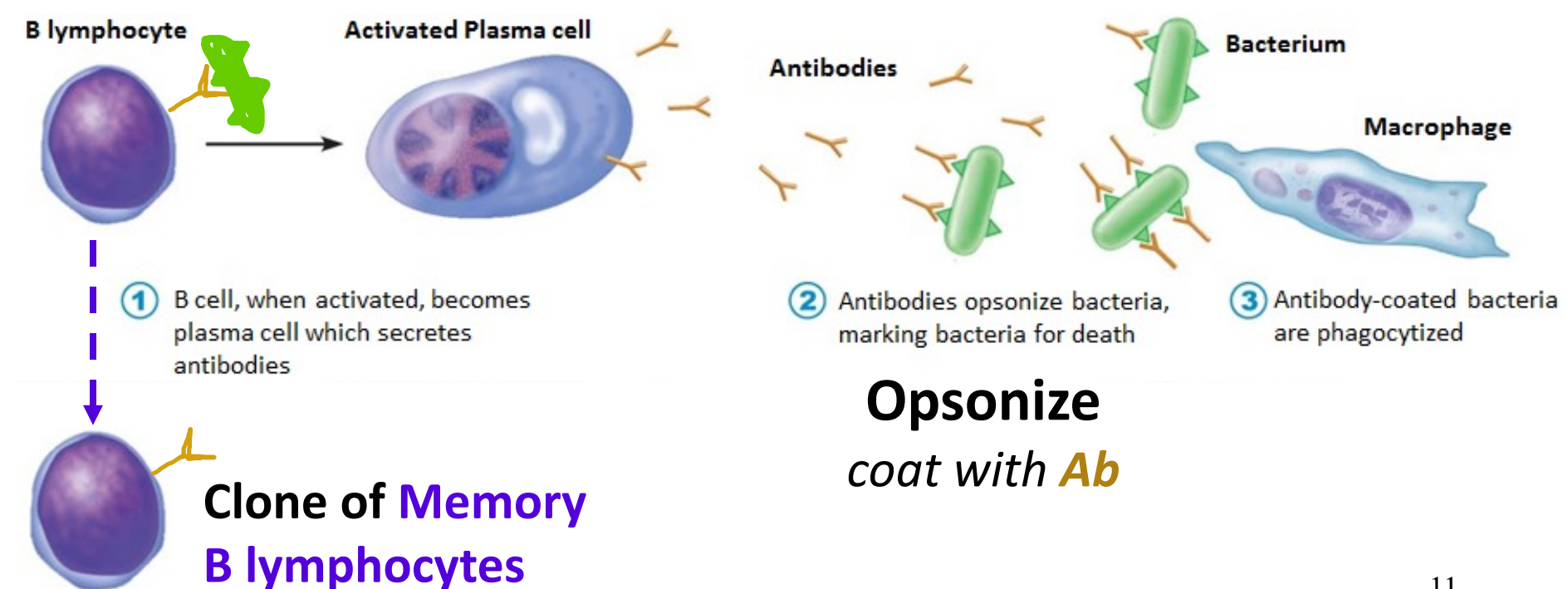
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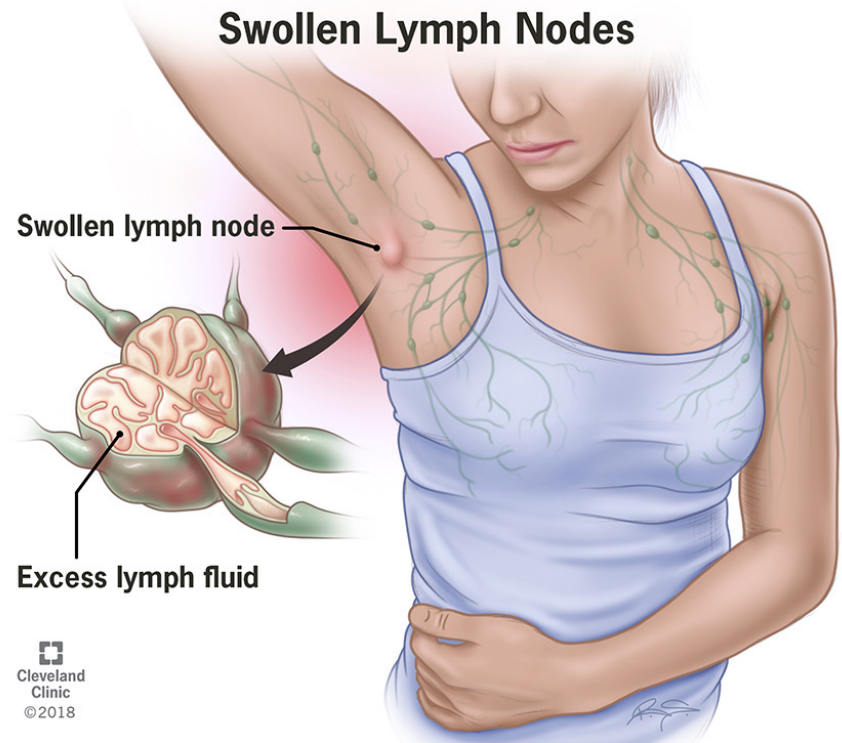
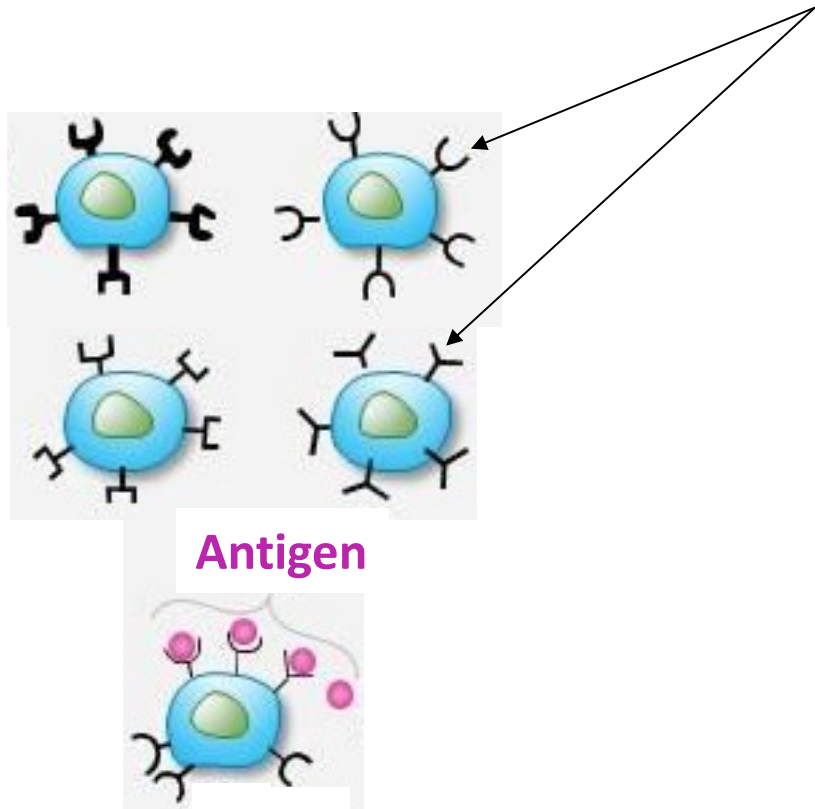
Mechanisms of Action

How do B lymphocytes cause humoral mediated immunity?

- B lymphocyte meets its specific **bacterial Ag** → is activated
- Activated B lymphocytes → plasma cells → **Ab's** → blood
- **Ab's** → bind to **Ag's** on **bacteria/free virus** → marks them for destruction
- **Memory B lymphocytes** remain; respond faster the next time

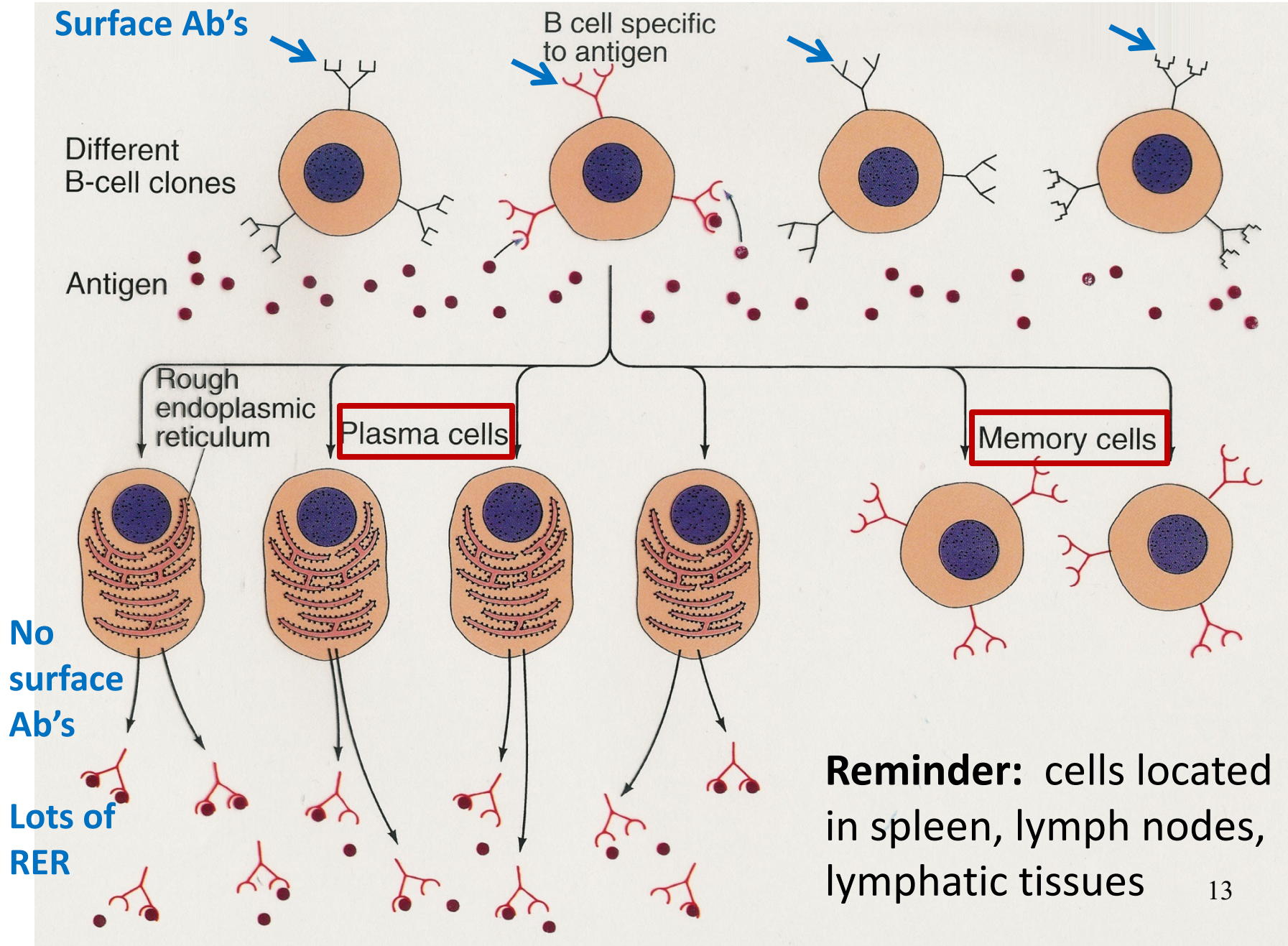


- Small **clones** of **B lymphocytes** live in lymphatic organs & tissues
- Marked with their unique **Ab** against a unique **pathogen**



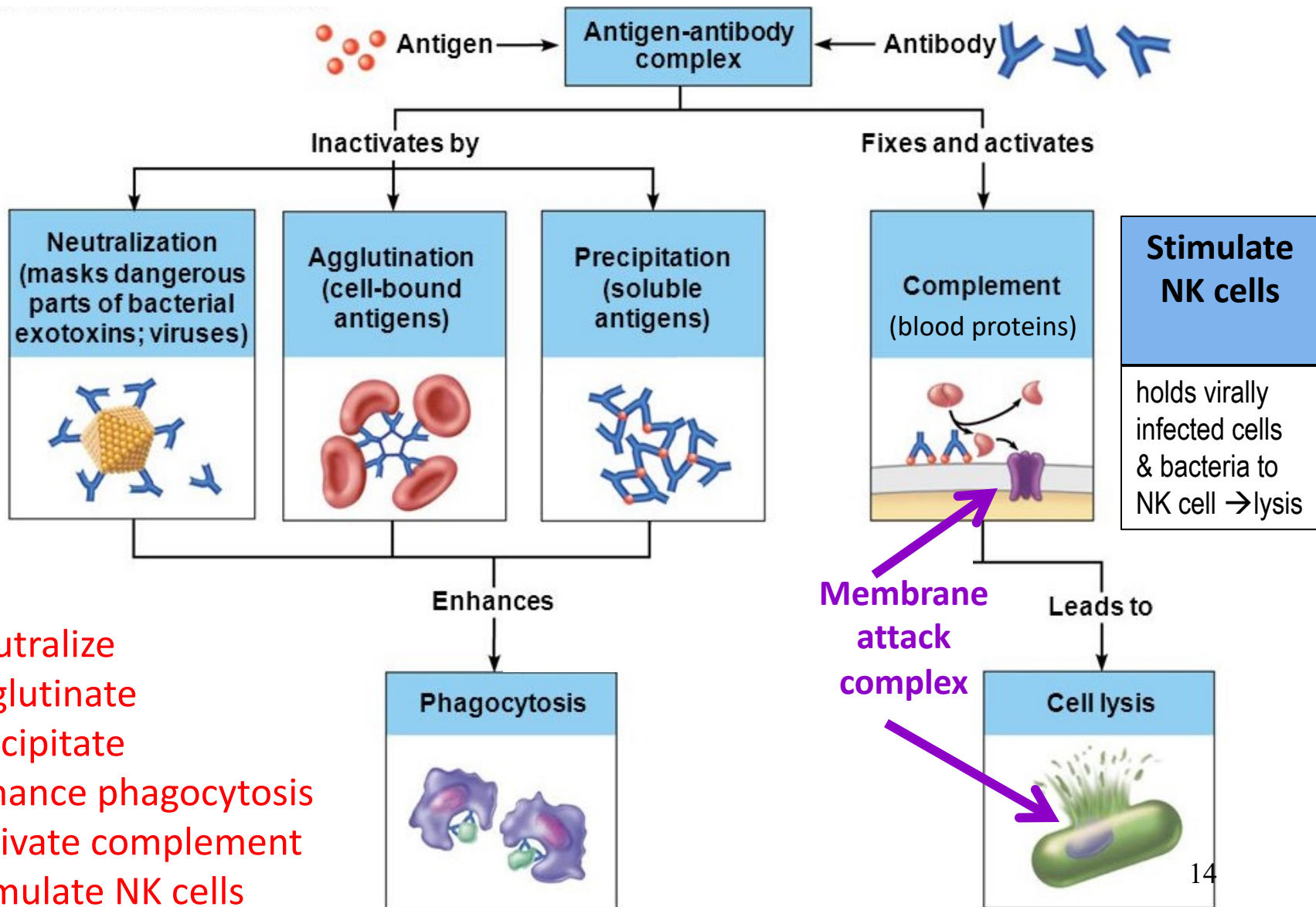
- Shows 5 unique **B lymphocyte clones**
- 1 is activated by **Ag** from **pathogen**
- Called **clonal selection**

How are clones selected? Clonal Selection Theory



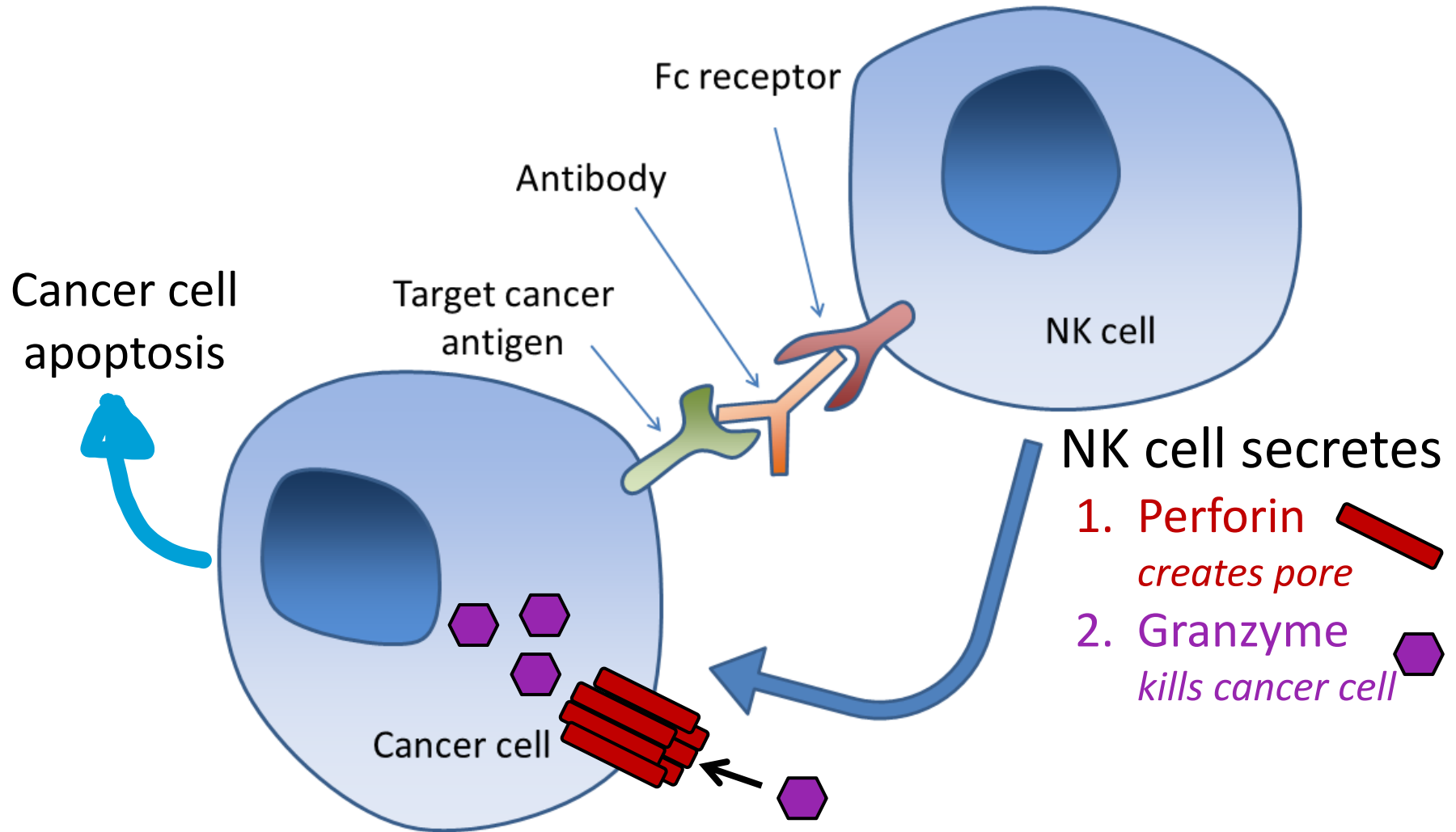
How do antibodies facilitate pathogen destruction?

binding of Ab-Ag alone does not destroy



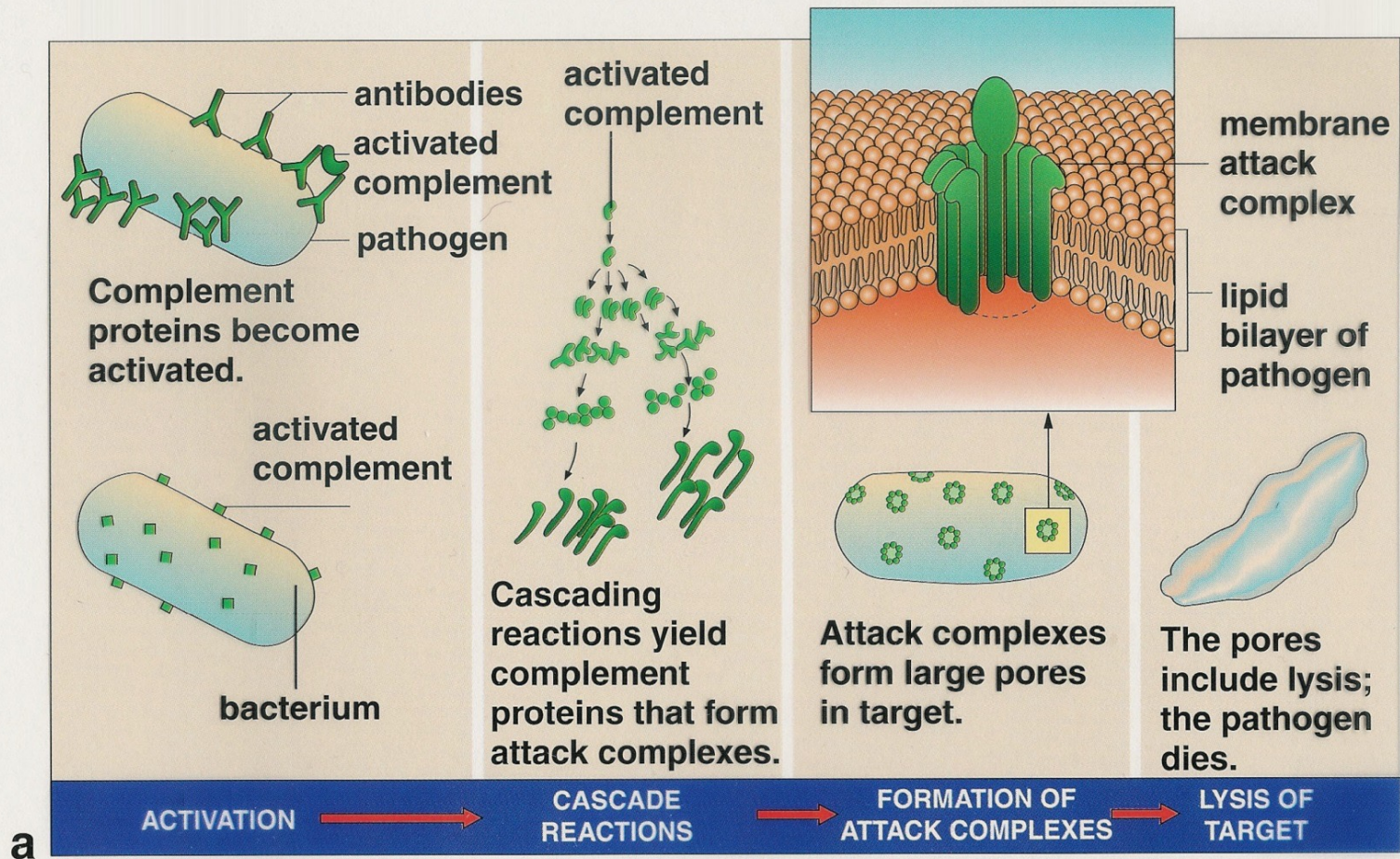
NK Cell Attack

Ab-Ag complex brings NK cell close to pathogen/cancer



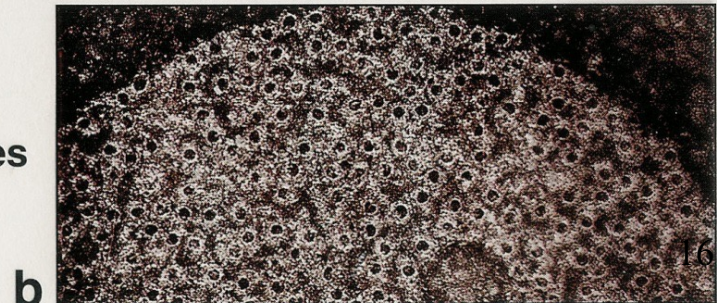
...or bacterium, free virus

Complement Activation → Membrane Attack Complex



Opsonize
coat with compliment

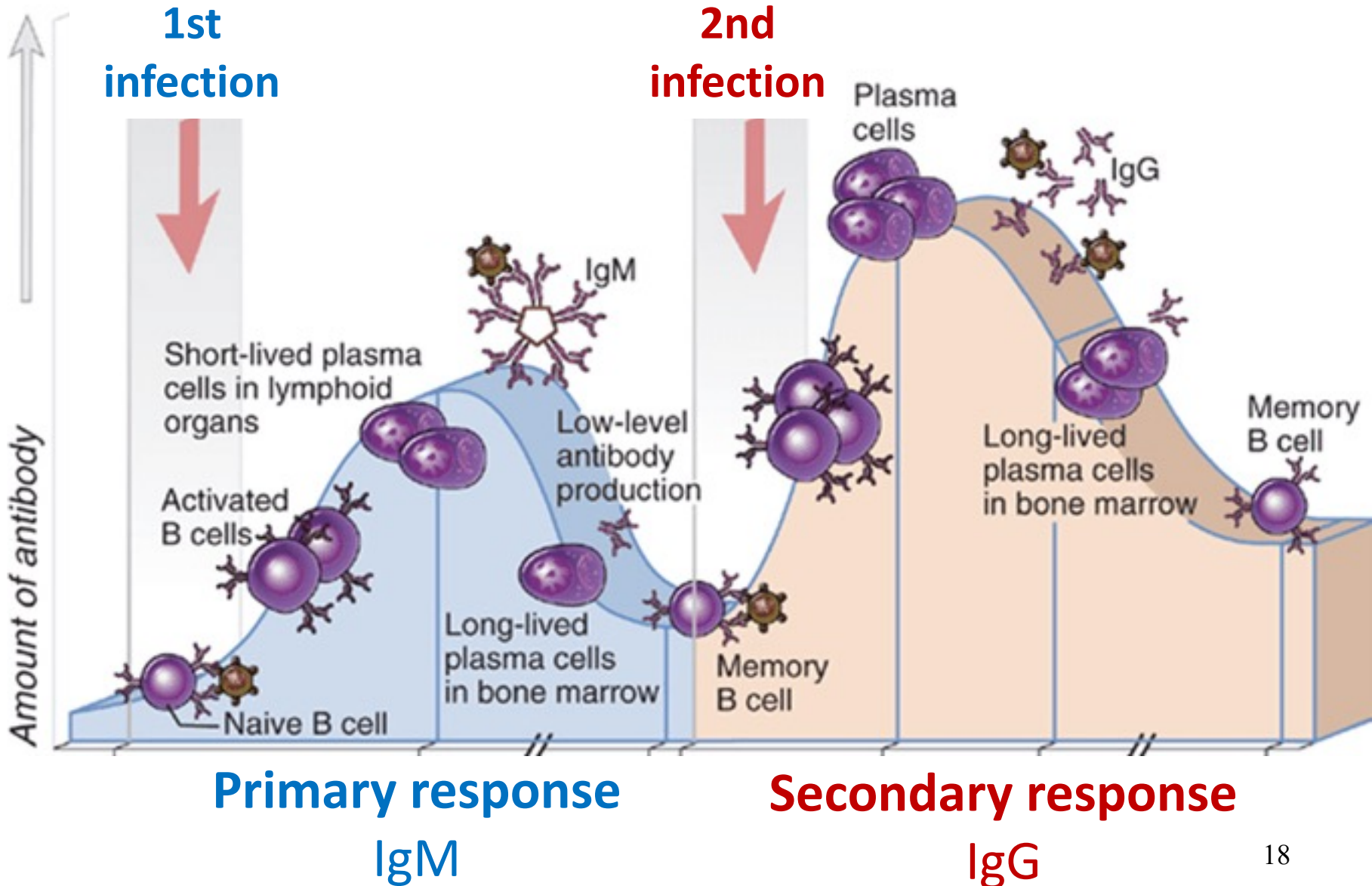
pore complexes



Classes of Antibodies

Class	Function
IgG	main form in circulation, increases after immunization “ <i>gamma globulin</i> ”; secreted in secondary response
IgA	main form in mucous membrane secretions (<i>milk, saliva, tears, mucus</i>)
IgE	mediate immediate hypersensitivity reactions (<i>allergies</i>); on mast cells
IgM	function as Ag receptor on B lymphocytes, secreted in primary response
IgD	function as Ag receptor on B lymphocytes, other functions unknown

Class switch from **IgM** → **IgG**



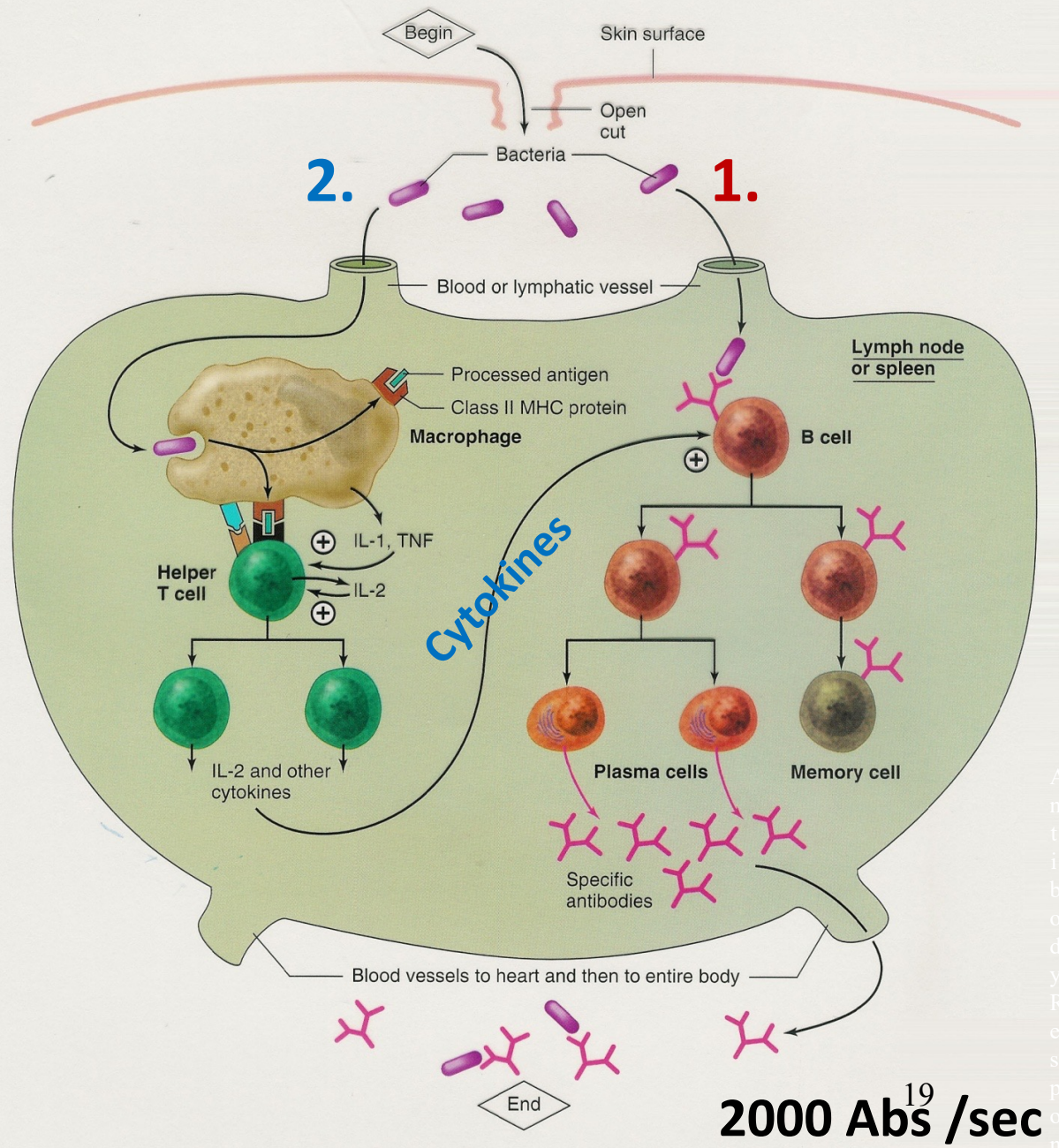
How do B cells get turned on?

B lymphocytes require dual activation:

1. Immune cell IDs pathogen Ag
chooses correct clone

2. Cytokines from helper T cells
enhance B lymphocyte response

B cell immune response to bacteria and free virus



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Specific Immune System

cell mediated immunity & cytotoxic T lymphocytes

T lymphocytes:

1. Cytotoxic T lymphocytes

- Secrete perforin & granzyme (*like NK cells*)
- Perforates and lyses **virally infected cells, cancer cells**

2. Helper T lymphocytes

- Secrete cytokines
- Help activate cytotoxic T cell response (& *B cell response*)

How do cytotoxic T cells get turned on?

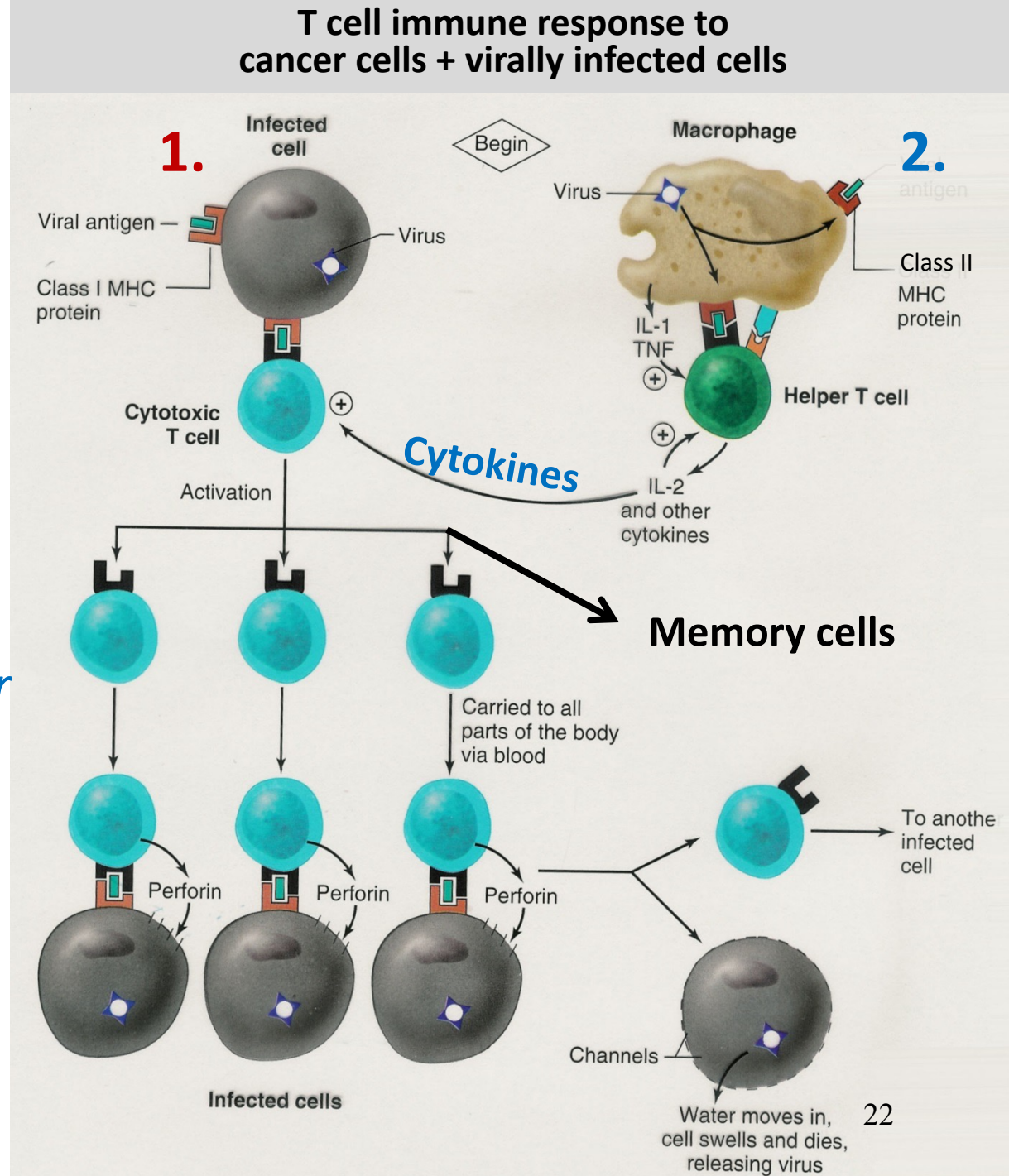
Cytotoxic T lymphocytes require **dual activation**:

1. Immune cell IDs

pathogen Ag
chooses correct clone

2. Cytokines from helper T cells

enhance cytotoxic T lymphocyte response



B Versus T lymphocytes

	B cell	helper T cell	cytotoxic T cell
maturation	bone marrow	thymus	thymus
receptor	IgM	TCR	TCR
recognition & activation <i>dual single</i>	antigen + T _H cell cytokines	antigen on class 2-MHC	antigen on class 1-MHC + T _H cell cytokines
response	secrete antibodies → <i>multiple outcomes</i>	secrete cytokines → <i>activate B cells + cytotoxic T cells</i>	secrete perforin → <i>lyse cells</i>
target	bacteria, free virus, bacterial toxins		virus-infected cells, cancer cells

Hallmarks of Specific Immune Response

Specificity – *how can B & T cells recognize antigens?*

receptors

Diversity – *how can B & T cells recognize different antigens?*

genetic recombination

Self / non-self – *how can B & T cells differentiate?*

clonal deletion; clonal inactivation

Memory – *how can B & T cells respond faster a second time?*

memory lymphocytes

Words to Know

Antigen

- Foreign molecule that initiates the specific immune response
- Binds to B cell antibody receptor or TCR

Antibodies

- Receptor and effector molecule of B cells
- Facilitates antigen destruction

Antibiotic

- Drug designed to kill/inhibit bacterial growth

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Primary & Secondary Immune Response

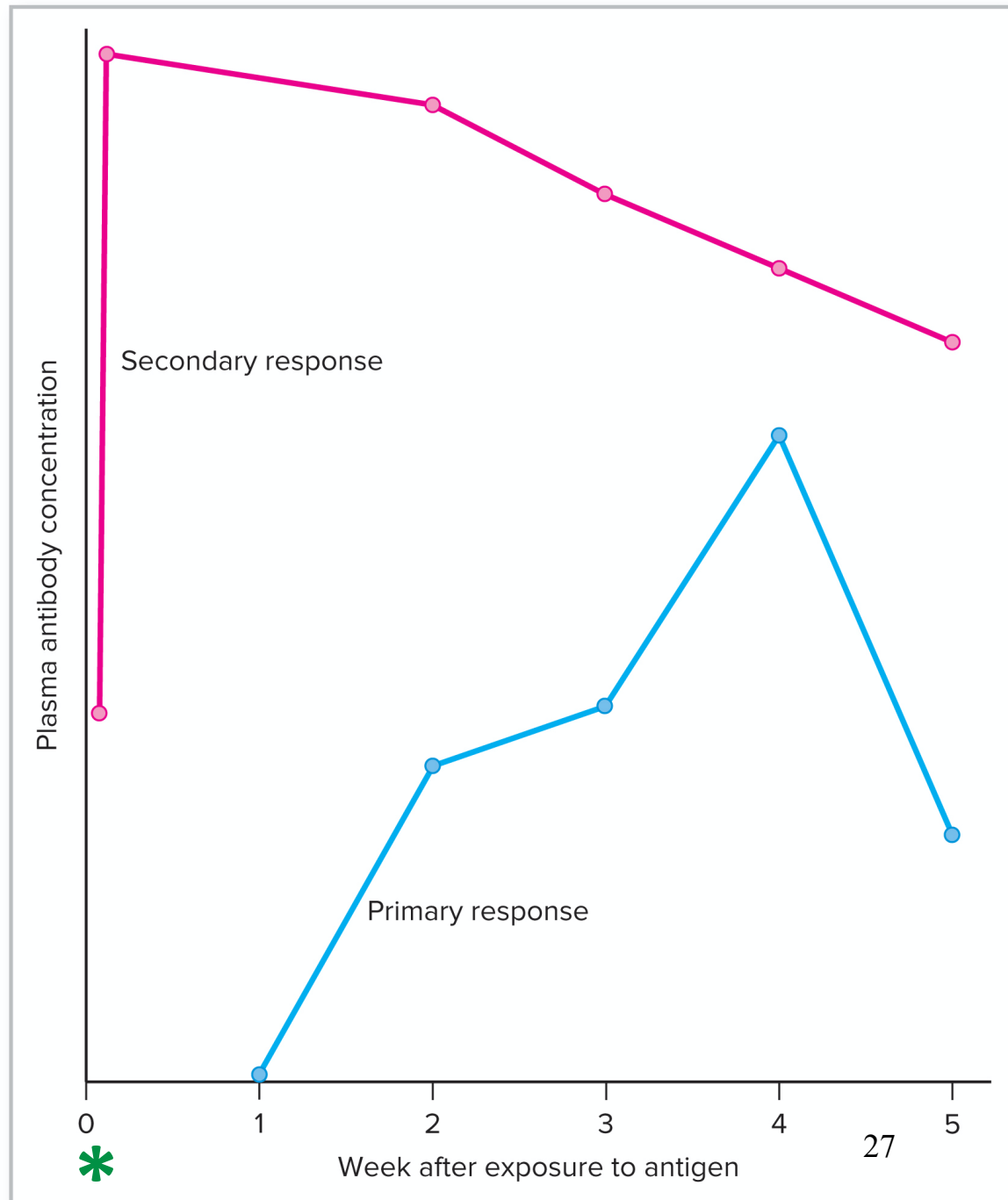
acquired immunity

- *Become ill*
- *Don't become ill*

Vaccines:

- Create *primary response without or with minimal illness*
- *Secondary response*
→ *less or no illness*

***Exposure to pathogen**



Acquisition of Immunity

mechanisms of immunization

	Natural acquired	Artificial acquired
Passive Borrowed from someone. <i>Short lasting</i>	Immunity acquired from antibodies passed in breast milk or through placenta * 	Immunity gained through antibodies harvested from another person or an animal * 
Active Created by exposure to Ag. <i>Long lasting</i>	Immunity gained through illness and recovery * 	Immunity acquired through a vaccine * 

*** Provides temporary protection through antibodies**

*** Provides immunity through memory cells**

Vaccine:

- Use attenuated pathogen or Ag, RND/DNA of Ag
- Used against bact. toxin, virus

How to Support Immunity

Support immunity:

1. Limit exposure – clean water, waste removal, wash hands, cough into arm
2. Increase resistance – healthy diet, exercise, adequate sleep
3. Protect most vulnerable – babies and seniors by limiting exposure, vaccination, breast milk

Suppress immunity:

- Poor nutrition, inadequate sleep, illness, stress